Magnetic Field in Matter

What is magnetism?

General answer: horse shoe magnet, compass needle, north pole

Loadstone

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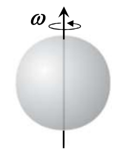
What is the connection of these with moving charges/current Carrying wires?

Obviously nothing…………at least macroscopically!

But all magnetic phenomenon are due to charges in motion!

What's going on? How about microscopically?



2

Magnetic Field in Matter

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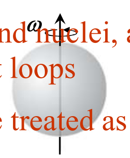
Electrons orbiting around nuclei, and spinning about their axis comprising tiny current loops



macroscopically can be treated as magnetic dipole !

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Magnetic Field in Matter

In general these dipoles are randomly oriented, and cancel each other.

With the application of the external magnetic field, individual dipole moments get aligned in the direction of the field

Though the alignment not complete………medium becomes magnetically polarized or MAGNETIZED

This phenomenon is called MAGNETIZATION

Magnetization parallel to **B** (Para-magnetization)

Magnetization anti parallel to **B** (Dia-magnetization)

Retains magnetization even after the removal of **B** (Ferro-magnetization)

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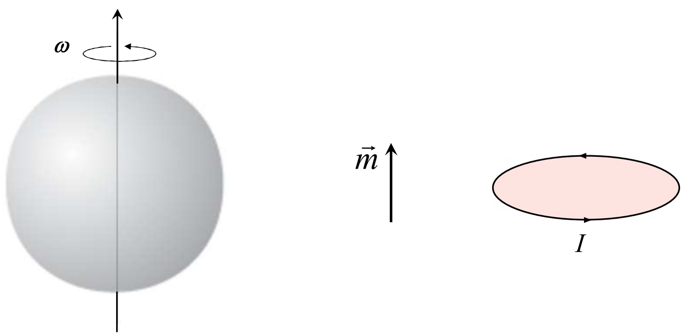
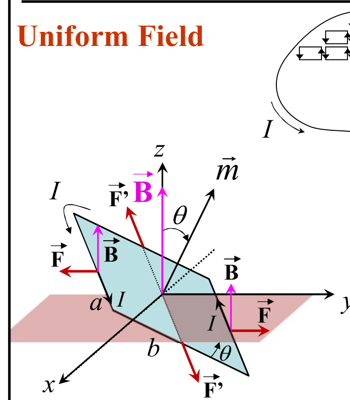
Magnetic Field in Matter

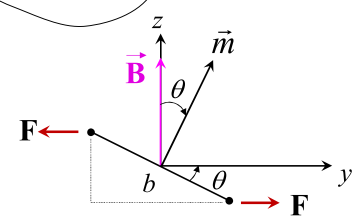
Effect of spinning electron in Uniform Field



Torque and force on magnetic dipole

Uniform Field





$$\vec{\mathbf{N}} = bF\sin\theta\,\hat{\mathbf{x}} \qquad F = IaB$$

$$\vec{\mathbf{N}} = IabB\sin\theta\,\hat{\mathbf{x}} = mB\sin\theta\,\hat{\mathbf{x}}$$

$$\boxed{\vec{\mathbf{N}} = \vec{\mathbf{m}} \times \vec{\mathbf{B}}}$$

$m = I\,a\,b$

Torque and force on magnetic dipole

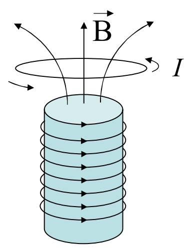
Uniform Field

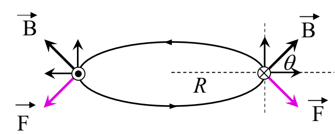
$$\boxed{\vec{\mathbf{N}} = \vec{\mathbf{m}} \times \vec{\mathbf{B}}}$$

- Force on the dipole: $\vec{\mathbf{F}} = I\oint d\vec{\mathbf{l}} \times \vec{\mathbf{B}} = I(\oint d\vec{\mathbf{l}}) \times \vec{\mathbf{B}} = \mathbf{0}$
- Tends to align the dipole in the direction of external field
- Responsible for Para-magnetism
- Every **spinning** electron in an atom constitute a magnetic dipole
- Is para-magnetism a universal phenomenon ?
- No. Essentially, you must have an atom/molecule with odd number of electrons

Torque and force on magnetic dipole

Non-Uniform Field





Net downward force: $\left|\vec{\mathbf{F}}\right| = \left|I\oint d\vec{\mathbf{l}} \times \vec{\mathbf{B}}\right| = I\oint dl\;B\cos\theta\sin 90^{\circ}$

$$= I\oint dl\;B\cos\theta \;= 2\pi RIB\cos\theta$$

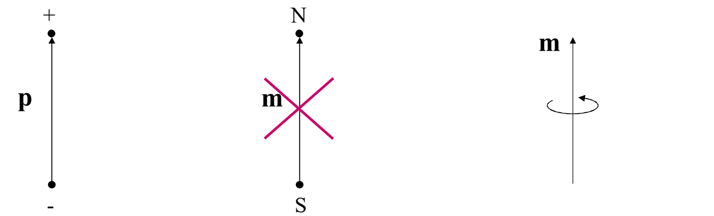
$$\boxed{\vec{\mathbf{F}} = \vec{\nabla}\left(\vec{\mathbf{m}} \cdot \vec{\mathbf{B}}\right)}$$

Torque and force on magnetic dipole

Uniform Field $\boxed{\vec{\mathbf{N}} = \vec{\mathbf{m}} \times \vec{\mathbf{B}}}$ $\boxed{\vec{\mathbf{F}} = 0}$

Non-Uniform Field $\boxed{\vec{\mathbf{N}} = \vec{\mathbf{m}} \times \vec{\mathbf{B}}}$ Exact torque about the center for a perfect dipole

$\boxed{\vec{\mathbf{F}} = \vec{\nabla}(\vec{\mathbf{m}} \cdot \vec{\mathbf{B}})}$



electric dipole   magnetic counterpart   real magnetic dipole

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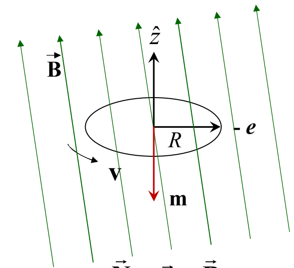
Effect of magnetic Field on Orbital motion

We analyzed effect of spinning electron. But electron also revolve around the nucleus $\rightarrow$ Orbital Motion



$\vec{\mathbf{N}} = \vec{\mathbf{m}} \times \vec{\mathbf{B}}$

$$T = \frac{2\pi R}{v}$$

$$I = \frac{e}{T} = \frac{e v}{2\pi R}$$

$$\vec{\mathbf{m}}\,(= I\,\pi R^2) = -\frac{1}{2} e\, v\, R\, \hat{\mathbf{z}}$$

Tends to align $\vec{m}$ along $\vec{B} \rightarrow$ possible only by tilting entire orbit !

Orbital contribution to Paramagnetism is extremely small.

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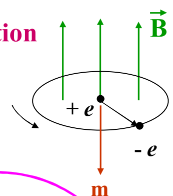
Effect of magnetic Field on Orbital motion

Lets take a deeper look of the effect of Orbital motion



Centripetal force $m_e \dfrac{v^2}{R} = \dfrac{1}{4\pi\varepsilon_0}\dfrac{e^2}{R^2}$

Consider $\vec{B}$ normal to the plane of the orbit !

$$m_e \frac{\bar{v}^2}{R} = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{R^2} + e\bar{v}B$$

inward $-e\,(\vec{\mathbf{v}} \times \vec{\mathbf{B}})$

$$e\cancel{\bar{v}}B = m_e\frac{\bar{v}^2}{R} - \frac{1}{4\pi\varepsilon_0}\frac{e^2}{R^2} = m_e\frac{\bar{v}^2}{R} - m_e\frac{v^2}{R}$$

$$= \frac{m_e}{R}\left(\bar{v}^2 - v^2\right) = \frac{m_e}{R}(\bar{v} - v)(\bar{v} + v) = \frac{m_e}{R}(\Delta v)\,2\cancel{\bar{v}}$$

Change in the Velocity: $\Delta v = \dfrac{eRB}{2m_e}$

With $\vec{B}$ on, electron speeds up!

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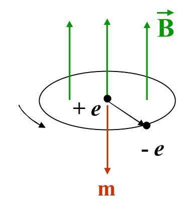
Effect of magnetic Field on Orbital motion

Change in the Velocity: $\Delta v = \dfrac{eRB}{2m_e}$



With $\vec{B}$ on, electron speeds up!

$\longrightarrow$ $\boxed{\Delta\,\vec{\mathbf{m}} = -\frac{1}{2} e\,(\Delta v)\,R\,\hat{\mathbf{z}} = -\frac{e^2 R^2}{4m_e}\vec{\mathbf{B}}}$

Change in $\vec{\mathbf{m}}$ is opposite to the direction of $\vec{\mathbf{B}}$!

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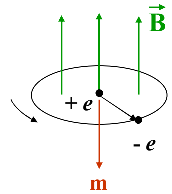
Effect of magnetic Field on Orbital motion

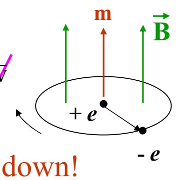
$$\vec{\mathbf{m}}\,(=I\,\pi\,R^2) = +\frac{1}{2}e\,\mathrm{v}\,R\,\hat{\mathbf{z}}$$

$$m_e\frac{\bar{\mathrm{v}}^2}{R} = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{R^2} - e\,\bar{\mathrm{v}}\,B \qquad -e\,(\vec{\mathrm{v}}\times\vec{\mathbf{B}}) \text{ outward}$$



$$e\cancel{\bar{\mathrm{v}}}\,B = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{R^2} - m_e\frac{\bar{\mathrm{v}}^2}{R} = m_e\frac{\mathrm{v}^2}{R} - m_e\frac{\bar{\mathrm{v}}^2}{R}$$

$$= \frac{m_e}{R}\left(\mathrm{v}^2-\bar{\mathrm{v}}^2\right) = \frac{m_e}{R}\left(\mathrm{v}-\bar{\mathrm{v}}\right)\left(\mathrm{v}+\bar{\mathrm{v}}\right) = -\frac{m_e}{R}\left(\Delta\,\mathrm{v}\right)2\cancel{\bar{\mathrm{v}}}$$



$\rightarrow$ $\Delta\,\mathrm{v} = -\dfrac{eRB}{2m_e}$ With $\vec{\mathrm{B}}$ on, electron slowed down!

$\rightarrow$ $\boxed{\Delta\,\vec{\mathbf{m}} = \frac{1}{2}e\left(\Delta\,\mathrm{v}\right)R\,\hat{\mathbf{z}} = -\frac{e^2R^2}{4m_e}\vec{\mathbf{B}}}$ Still, change in $\vec{\mathbf{m}}$ is opposite to the direction of $\vec{\mathbf{B}}$!

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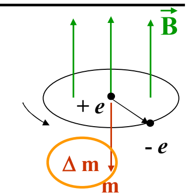
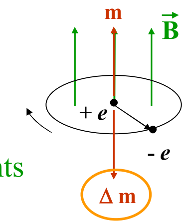
Effect of magnetic Field on Orbital motion

- With $\vec{\mathbf{B}}$ on, electron speeds up!
- Change in $\vec{\mathbf{m}}$ is opposite to the direction of $\vec{\mathbf{B}}$!



- With $\vec{\mathbf{B}}$ on, electron slowed down!
- Still, change in $\vec{\mathbf{m}}$ is opposite to the direction of $\vec{\mathbf{B}}$!



In general, electron orbits are randomly oriented, canceling individual dipole moments

With $\vec{\mathrm{B}}$ on, each atom picks up little "extra" **dipole moment increment** anti-parallel to $\vec{\mathrm{B}}$ **Responsible for Diamagnetism**

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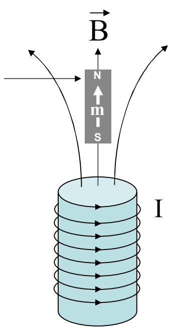
Magnetic material in external field

Paramagnetic material



Induced dipole moment $\rightarrow$ upward

Force on the object $\rightarrow$ downward

Paramagnet is attracted into the field

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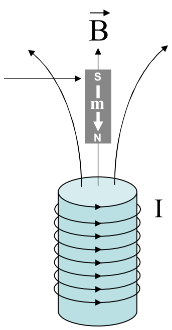
Magnetic material in external field

Diamagnetic material



Induced dipole moment $\rightarrow$ downward

Force on the object $\rightarrow$ upward

Diamagnet is repelled away by the field

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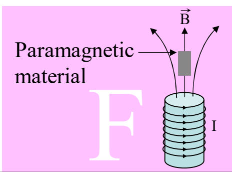
Magnetic material in external field

Strength of diamagnetism (alteration of orbital speed in such a way to give rise the change in orbital dipole moment in the direction opposite to the external field)

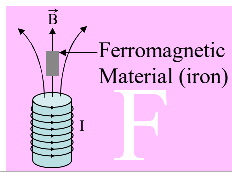
much weaker than

Strength of paramagnetism (torque on dipole associated with the spin of unpaired electron to align dipole parallel to the external field)

In general, diamagnetism and paramagnetism are extremely weak phenomenon



$= 10^{-4}/10^{-5} \times$



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Magnetic material in external field - Conclusion

With the application of the external magnetic field, medium becomes magnetically polarized or MAGNETIZED

This phenomenon is called MAGNETIZATION

Magnetization: $\vec{\mathbf{M}}$ = *magnetic dipole per unit volume*

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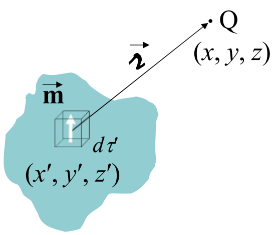
Field of Magnetized Object



Magnetized material

$$\vec{\mathbf{A}}(\vec r) = \frac{\mu_0}{4\pi}\frac{\vec m \times \hat r}{r^2}$$

$$\mathbf{d}\vec{\mathbf{m}}(\mathbf{r}') = \vec{\mathbf{M}}(\mathbf{r}')\, d\tau'$$

$$\vec{\mathbf{A}}(\vec{\mathbf{r}}) = \frac{\mu_0}{4\pi}\int \frac{\vec{\mathbf{M}}(\mathbf{r}') \times \hat{\mathscr{r}}}{\mathscr{r}^2}\, d\tau'$$

$$\mathscr{r} = [(x-x')^2 + (y-y')^2 + (z-z')^2]^{1/2}$$

$$\vec\nabla'\left(\frac{1}{\mathscr{r}}\right) = \frac{\hat{\mathscr{r}}}{\mathscr{r}^2}$$

$$\vec{\mathbf{A}}(\vec{\mathbf{r}}) = \frac{\mu_0}{4\pi}\int \vec{\mathbf{M}}(\mathbf{r}') \times \vec\nabla'\left(\frac{1}{\mathscr{r}}\right) d\tau'$$

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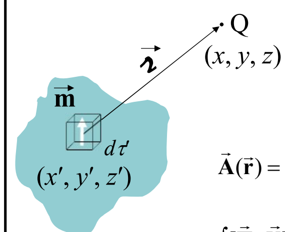
Field of Magnetized Object



Magnetized material

$$\vec{\mathbf{A}}(\vec{\mathbf{r}}) = \frac{\mu_0}{4\pi}\int \vec{\mathbf{M}}(\mathbf{r}') \times \vec\nabla'\left(\frac{1}{\mathscr{r}}\right) d\tau'$$

$$\vec\nabla \times \left(f\vec{\mathbf{A}}\right) = f\left(\vec\nabla \times \vec{\mathbf{A}}\right) - \vec{\mathbf{A}} \times \left(\vec\nabla f\right)$$

$$\vec{\mathbf{A}}(\vec{\mathbf{r}}) = \frac{\mu_0}{4\pi}\left[\int \frac{1}{\mathscr{r}}[\vec\nabla' \times \vec{\mathbf{M}}(\mathbf{r}')]\, d\tau' - \int \vec\nabla' \times \left(\frac{\vec{\mathbf{M}}(\mathbf{r}')}{\mathscr{r}}\right) d\tau'\right]$$

$$\int_V [\vec\nabla \cdot \vec V]\, d\tau = \oint_S \vec V \cdot d\vec a$$

$$\int_V [\vec\nabla \cdot (\vec c \times \vec V)]\, d\tau = \oint_S (\vec c \times \vec V) \cdot d\vec a$$

$$\int_V [\vec V \cdot (\cancel{\vec\nabla \times \vec c}) - \vec c \cdot (\vec\nabla \times \vec V)]\, d\tau = \oint_S d\vec a \cdot (\vec c \times \vec V)$$

$$-\int_V \vec c \cdot (\vec\nabla \times \vec V)\, d\tau = \oint_S \vec c \cdot (\vec V \times d\vec a) \Rightarrow -\int_V (\vec\nabla \times \vec V)\, d\tau = \oint_S (\vec V \times d\vec a)$$

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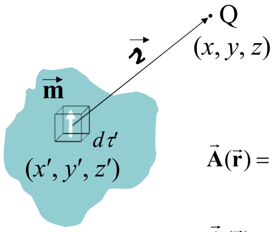
Field of Magnetized Object



Magnetized material

$$\vec{\mathbf{A}}(\vec{\mathbf{r}}) = \frac{\mu_0}{4\pi}\int \vec{\mathbf{M}}(\mathbf{r}') \times \vec{\nabla}'\left(\frac{1}{\mathscr{r}}\right) d\tau'$$

$$\vec{\nabla} \times \left(f\vec{\mathbf{A}}\right) = f\left(\vec{\nabla} \times \vec{\mathbf{A}}\right) - \vec{\mathbf{A}} \times \left(\vec{\nabla} f\right)$$

$$\vec{\mathbf{A}}(\vec{\mathbf{r}}) = \frac{\mu_0}{4\pi}\left[\int \frac{1}{\mathscr{r}}[\vec{\nabla}' \times \vec{\mathbf{M}}(\mathbf{r}')]d\tau' - \int \vec{\nabla}' \times \left(\frac{\vec{\mathbf{M}}(\mathbf{r}')}{\mathscr{r}}\right) d\tau'\right]$$

$$\vec{\mathbf{A}}(\vec{\mathbf{r}}) = \frac{\mu_0}{4\pi}\int \frac{1}{\mathscr{r}}[\vec{\nabla}' \times \vec{\mathbf{M}}(\mathbf{r}')]d\tau' + \frac{\mu_0}{4\pi}\oint_S \frac{1}{\mathscr{r}}[\vec{\mathbf{M}}(\mathbf{r}') \times d\vec{\mathbf{a}}]$$

$$-\int_V \vec{c} \cdot (\vec{\nabla} \times \vec{V})\, d\tau = \oint_S \vec{c} \cdot (\vec{V} \times d\vec{a}) \Rightarrow -\int_V (\vec{\nabla} \times \vec{V})\, d\tau = \oint_S (\vec{V} \times d\vec{a})$$

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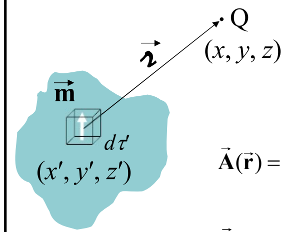
Field of Magnetized Object



Magnetized material

$$\vec{\mathbf{A}}(\vec{\mathbf{r}}) = \frac{\mu_0}{4\pi}\int \vec{\mathbf{M}}(\mathbf{r}') \times \vec{\nabla}'\left(\frac{1}{\mathscr{r}}\right) d\tau'$$

$$\vec{\nabla} \times \left(f\vec{\mathbf{A}}\right) = f\left(\vec{\nabla} \times \vec{\mathbf{A}}\right) - \vec{\mathbf{A}} \times \left(\vec{\nabla} f\right)$$

$$\vec{\mathbf{A}}(\vec{\mathbf{r}}) = \frac{\mu_0}{4\pi}\left[\int \frac{1}{\mathscr{r}}[\vec{\nabla}' \times \vec{\mathbf{M}}(\mathbf{r}')]d\tau' - \int \vec{\nabla}' \times \left(\frac{\vec{\mathbf{M}}(\mathbf{r}')}{\mathscr{r}}\right) d\tau'\right]$$

$$\vec{\mathbf{A}}(\vec{\mathbf{r}}) = \frac{\mu_0}{4\pi}\int \frac{1}{\mathscr{r}}[\vec{\nabla}' \times \vec{\mathbf{M}}(\mathbf{r}')]d\tau' + \frac{\mu_0}{4\pi}\oint_S \frac{1}{\mathscr{r}}[\vec{\mathbf{M}}(\mathbf{r}') \times d\vec{\mathbf{a}}]$$

$$\vec{\mathbf{A}}(\mathbf{r}) = \frac{\mu_0}{4\pi}\int \frac{\mathbf{J}(\mathbf{r}')}{\mathscr{r}} d\tau'$$

$$\vec{\mathbf{A}}(\mathbf{r}) = \frac{\mu_0}{4\pi}\int \frac{\mathbf{K}}{\mathscr{r}} da'$$

Potential due to volume current

Potential due to surface current

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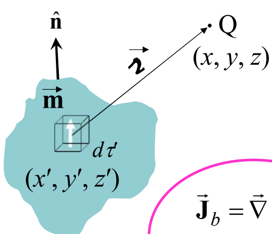
Field of Magnetized Object



Magnetized material

$$\vec{\mathbf{A}}(\vec{\mathbf{r}}) = \frac{\mu_0}{4\pi}\int \frac{1}{\mathscr{r}}[\vec{\nabla}' \times \vec{\mathbf{M}}(\mathbf{r}')]d\tau' + \frac{\mu_0}{4\pi}\oint_S \frac{1}{\mathscr{r}}[\vec{\mathbf{M}}(\mathbf{r}') \times d\vec{\mathbf{a}}]$$

$$\vec{\mathbf{A}}(\vec{\mathbf{r}}) = \frac{\mu_0}{4\pi}\int \frac{\vec{\mathbf{J}}_b(\vec{\mathbf{r}}')}{\mathscr{r}} d\tau' + \frac{\mu_0}{4\pi}\oint_S \frac{\vec{\mathbf{K}}_b(\vec{\mathbf{r}}')}{\mathscr{r}} da'$$

$\vec{\mathbf{J}}_b = \vec{\nabla} \times \vec{\mathbf{M}}$ **Bound volume current density**

$\vec{\mathbf{K}}_b = \vec{\mathbf{M}} \times \hat{\mathbf{n}}$ **Bound surface current density**

$\rho_b = -\vec{\nabla} \cdot \vec{\mathrm{P}}$ $\qquad \sigma_b = \vec{\mathrm{P}} \cdot \hat{\mathrm{n}}$

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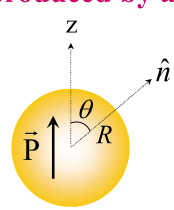
Example *from electrostatics (Example 4.2) to recall*

$\vec{\mathrm{E}}$ produced by a uniformly polarized sphere of radius *R*



r < R

$$\vec{\mathrm{E}} = -\vec{\nabla} V = -\frac{P}{3\varepsilon_0}\hat{z} = -\frac{1}{3\varepsilon_0}\vec{\mathrm{P}}$$

Field is constant inside and $\uparrow\downarrow \vec{\mathrm{P}}$

r > R

$$V = \frac{1}{4\pi\varepsilon_0}\frac{\vec{p} \cdot \hat{r}}{r^2}$$

Potential of a perfect dipole at the origin

$$\vec{p} = \frac{4}{3}\pi R^3\, \vec{\mathrm{P}}$$

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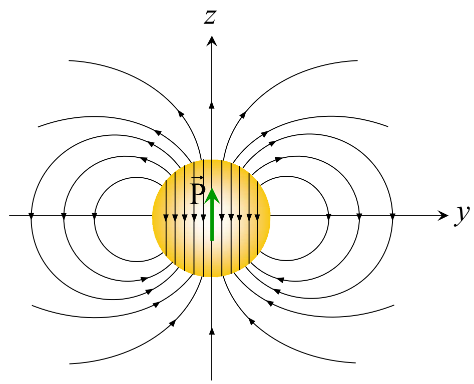
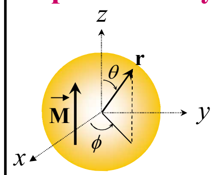
Example *from electrostatics (Example 4.2) to recall*

$\vec{\mathrm{E}}$ produced by a uniformly polarized sphere of radius R



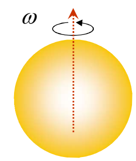
Example *for magnetostatics*

$\vec{\mathrm{B}}$ produced by a uniformly magnetized sphere of radius R



$$\vec{\mathbf{J}}_b = \vec{\nabla} \times \vec{\mathbf{M}} = \mathbf{0}$$

$$\vec{\mathbf{K}}_b = \vec{\mathbf{M}} \times \hat{\mathbf{n}} = M \sin\theta\, \hat{\phi}$$



A spinning spherical shell of uniform surface density σ

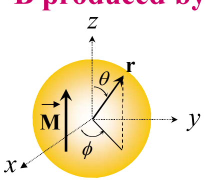
$\rightarrow$ Surface current density $\vec{\mathbf{K}} = \sigma\, \vec{\mathbf{v}} = \sigma\, R\omega \sin\theta\, \hat{\boldsymbol{\varphi}}$

$\sigma R\, \vec{\omega} \rightarrow \vec{\mathbf{M}}$

Field of uniformly magnetized sphere is identical to the field of a spinning spherical shell

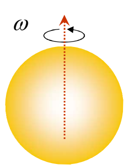
Example *for magnetostatics*

$\vec{\mathrm{B}}$ produced by a uniformly magnetized sphere of radius R



$$\vec{\mathbf{J}}_b = \vec{\nabla} \times \vec{\mathbf{M}} = 0$$

$$\vec{\mathbf{K}}_b = \vec{\mathbf{M}} \times \hat{\mathbf{n}} = M \sin\theta\, \hat{\phi}$$



A spinning spherical shell of uniform surface density σ

$\rightarrow$ Surface current density $\vec{\mathbf{K}} = \sigma\, \vec{\mathbf{v}} = \sigma\, R\omega \sin\theta\, \hat{\boldsymbol{\varphi}}$

$\sigma R\, \vec{\omega} \rightarrow \vec{\mathbf{M}}$

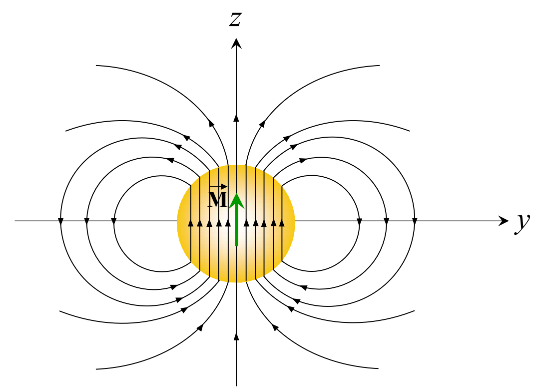
inside $\vec{\mathbf{B}} = \frac{2}{3}\mu_0 \sigma R \vec{\omega} = \frac{2}{3}\mu_0 \vec{\mathbf{M}}$ Uniform [Ex:5.11]

Field of uniformly magnetized sphere is identical to the field of a spinning spherical shell

outside $\vec{\mathbf{m}} = \frac{4}{3}\pi \sigma R \vec{\omega} R^3 = \frac{4}{3}\pi R^3 \vec{\mathbf{M}}$ pure dipole [HW]

Example *for magnetostatics*

$\vec{\mathrm{B}}$ produced by a uniformly magnetized sphere of radius R



$\vec{\mathbf{H}}$

$$\vec{\mathbf{J}} = \vec{\mathbf{J}}_b + \vec{\mathbf{J}}_f \qquad \vec{\nabla} \times \vec{\mathbf{B}} = \mu_0 \vec{\mathbf{J}}$$

$$\frac{1}{\mu_0}(\vec{\nabla} \times \vec{\mathbf{B}}) = \vec{\mathbf{J}} = \vec{\mathbf{J}}_b + \vec{\mathbf{J}}_f = \vec{\mathbf{J}}_f + \vec{\nabla} \times \vec{\mathbf{M}}$$

$$\vec{\nabla} \times \left(\frac{1}{\mu_0} \vec{\mathbf{B}} - \vec{\mathbf{M}} \right) = \vec{\mathbf{J}}_f$$

$$\boxed{\vec{\mathbf{H}} = \left(\frac{1}{\mu_0} \vec{\mathbf{B}} - \vec{\mathbf{M}} \right)}$$

$$\vec{\nabla} \times \vec{\mathbf{H}} = \vec{\mathbf{J}}_f$$

$$\oint \vec{\mathbf{H}} \cdot d\vec{\mathbf{l}} = I_{f_{enc}}$$

Ampere's Law for $\vec{\mathbf{H}}$

Apply if $\vec{\mathbf{B}}$ or $\vec{\mathbf{H}}$ needed when there is high degree of symmetry

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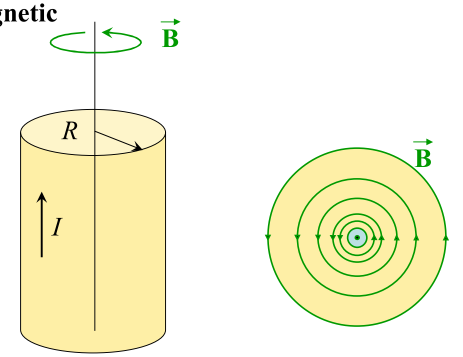
Example

Long copper rod with uniformly distributed (free) current I

$\vec{\mathbf{H}}$ (inside and outside)?

Copper is weakly diamagnetic



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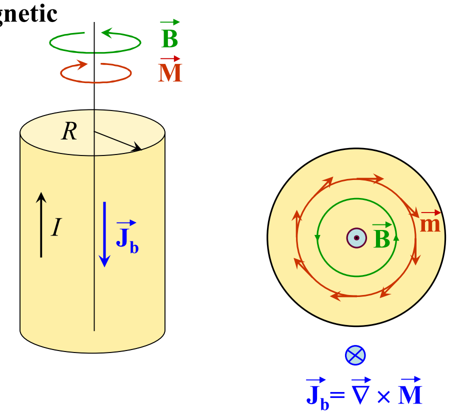
Example

Long copper rod with uniformly distributed (free) current I

$\vec{\mathbf{H}}$ (inside and outside)?

Copper is weakly diamagnetic



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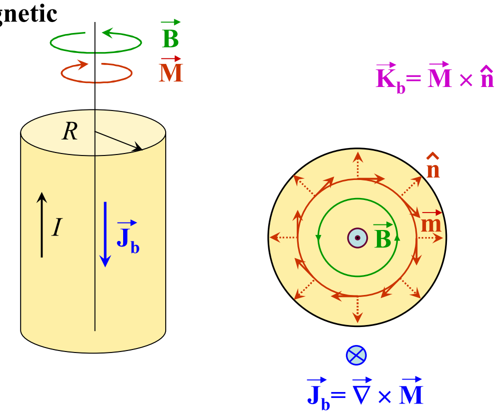
Example

Long copper rod with uniformly distributed (free) current I

$\vec{\mathbf{H}}$ (inside and outside)?

Copper is weakly diamagnetic



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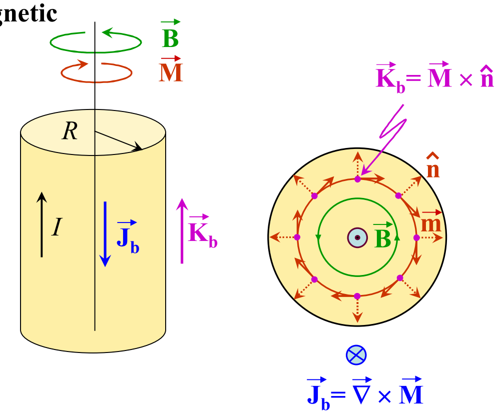
Example

Long copper rod with uniformly distributed (free) current I
$\vec{H}$ (inside and outside)?

Copper is weakly diamagnetic

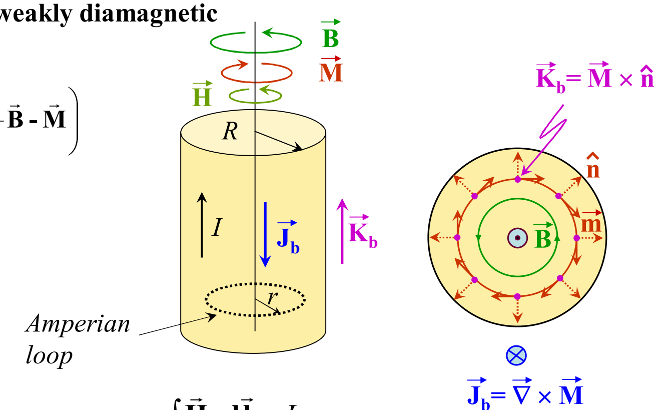


Example

Long copper rod with uniformly distributed (free) current I
$\vec{H}$ (inside and outside)?

Copper is weakly diamagnetic

$$\vec{H} = \left(\frac{1}{\mu_0}\vec{B} - \vec{M}\right)$$



H. E.

$$\oint \vec{H}\cdot d\vec{l} = I_{f_{enc}}$$

Constituting Relation

$\vec{M} = \chi_m \vec{H}$

→ **Magnetic Susceptibility** **(dimensionless constant)**

- → **+ ve paramagnets**
- → **– ve diamagnets**
- → **>> 0 ferromagnets**

$\vec{H} = \left(\frac{1}{\mu_0}\vec{B} - \vec{M}\right)$

$\vec{B} = \mu_0\left(\vec{H} + \vec{M}\right)$

$\vec{B} = \mu_0\left(\vec{H} + \chi_m\vec{H}\right)$

Permeability of the material

For vacuum: $\chi_m = 0$

$\vec{B} = \mu_0(1+\chi_m)\vec{H}$ ➡ $\boxed{\vec{B} = \mu\vec{H}}$ ➡ $\boxed{\mu = \mu_0(1+\chi_m)}$ $\quad \frac{\mu}{\mu_0} = (1+\chi_m) = \mu_r$

Medium

$\chi_m \neq \chi_m(\vec{r})$		**Homogeneous Medium**
$\chi_m = \chi_m(\vec{r})$		**Inhomogeneous Medium**
$\chi_m \neq \chi_m(\vec{H}, \vec{r})$	$\vec{M} = \chi_m \vec{H}$ $\vec{B} = \mu\vec{H}$	**Linear Medium** paramagnets diamagnets
$\chi_m = \chi_m(\vec{H})$	$\cancel{\vec{M} = \chi_m \vec{H}}$ $\cancel{\vec{B} = \mu\vec{H}}$	**Non-Linear Medium** ferromagnets
$\chi_m = $ Constant	$\vec{M} \parallel \vec{H}$ $\vec{B} \parallel \vec{H}$	**Isotropic Medium**
$\chi_m = $ tensor	$\vec{M} \nparallel \vec{H}$ $\vec{B} \nparallel \vec{H}$	**Non-Isotropic Medium**

Boundary Condition

$\vec{\nabla} \cdot \vec{\mathbf{B}} = 0$

$B^{\perp}_{\text{above}} = B^{\perp}_{\text{below}}$

Continuous

$\Rightarrow$ $\vec{\mathbf{B}} = \mu\, \vec{\mathbf{H}}$

$\mu_{\text{above}}\, H^{\perp}_{\text{above}} = \mu_{\text{below}}\, H^{\perp}_{\text{below}}$

Discontinuous

$\vec{\nabla} \times \vec{\mathbf{H}} = \vec{\mathbf{J}}_f$

$H^{||}_{above} - H^{||}_{below} = K_f$

Discontinuous

$\Rightarrow$ $\dfrac{B^{||}_{\text{above}}}{\mu_{\text{above}}} - \dfrac{B^{||}_{\text{below}}}{\mu_{\text{below}}} = K_f$

Discontinuous

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Boundary Condition

In the absence of any free current

$\vec{\nabla} \cdot \vec{\mathbf{B}} = 0$

$B^{\perp}_{\text{above}} = B^{\perp}_{\text{below}}$

Continuous

$\Rightarrow$ $\vec{\mathbf{B}} = \mu\, \vec{\mathbf{H}}$

$\mu_{\text{above}}\, H^{\perp}_{\text{above}} = \mu_{\text{below}}\, H^{\perp}_{\text{below}}$

Discontinuous

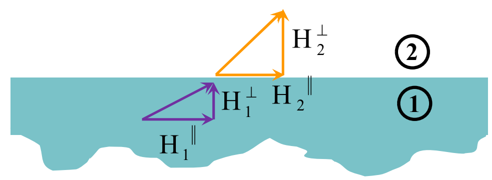
$\vec{\nabla} \times \vec{\mathbf{H}} = 0 \quad \vec{\mathbf{H}} = -\vec{\nabla} V_{\text{mag}}$

$H^{||}_{above} = H^{||}_{below}$

Continuous

$\Rightarrow$ $\mu_{\text{below}} B^{||}_{\text{above}} = \mu_{\text{above}} B^{||}_{\text{below}}$

Discontinuous



$H_2^{\perp} = \dfrac{\mu_1}{\mu_2} H_1^{\perp}$

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Boundary Condition

In the absence of any free current

- $B^{\perp}$ is always continuous across the boundary
- $H^{\perp}$ is discontinuous across the boundary
- $B^{||}$ is discontinuous across the boundary
- $H^{||}$ is always continuous across the boundary

- $D^{\perp}$ is always continuous across the boundary
- $E^{\perp}$ is discontinuous across the boundary
- $D^{||}$ is discontinuous across the boundary
- $E^{||}$ is always continuous across the boundary

$\vec{D}$ follows BC as was by $\vec{B}$ and $\vec{E}$ follows BC as was by $\vec{H}$

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Boundary Condition

In the absence of any free charge

$\vec{\nabla} \cdot \vec{\mathbf{D}} = 0 \qquad \vec{\nabla} \times \vec{\mathbf{E}} = 0 \qquad \varepsilon_0 \vec{\mathbf{E}} = \vec{\mathbf{D}} - \vec{\mathbf{P}}$

In the absence of any free current

$\vec{\nabla} \cdot \vec{\mathbf{B}} = 0 \qquad \vec{\nabla} \times \vec{\mathbf{H}} = 0 \qquad \mu_0 \vec{\mathbf{H}} = \vec{\mathbf{B}} - \mu_0 \vec{\mathbf{M}}$

Replacing $\vec{D}$ to $\vec{B}$, $\vec{E}$ to $\vec{H}$, $\vec{P}$ to $\mu_0\vec{M}$ and ε_0 to μ_0

Electrostatic Equations $\rightarrow$ Magnetostatic Equations

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Boundary Condition

In the absence of any free current

This means, we have already solved so many magnetostatics problem

$$\vec{\mathbf{B}} = \mu\,\vec{\mathbf{H}}$$
$$\vec{\mathbf{D}} = \varepsilon\,\vec{\mathbf{E}}$$

Replacing $\vec{D}$ to $\vec{B}$, $\vec{E}$ to $\vec{H}$, $\vec{P}$ to $\vec{\mu_0 M}$ and ε_0 to μ_0

Electrostatic Equations $\rightarrow$ Magnetostatic Equations

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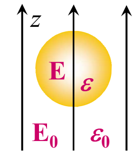
Boundary Condition

Example: 1

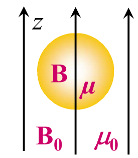
A sphere of homogeneous linear dielectric material is placed in an otherwise uniform field E_0. Find the dipole moment of the sphere.



$$\boxed{\vec{\mathbf{p}} = \left(\frac{\varepsilon - \varepsilon_0}{\varepsilon + 2\varepsilon_0}\right)\vec{\mathbf{E}}_0 R^3}$$

Example: 2

A sphere of homogeneous linear magnetic material is placed in an otherwise uniform field B_0. Find the dipole moment of the sphere.



$$\boxed{\vec{\mathbf{m}} = \left(\frac{\mu - \mu_0}{\mu + 2\mu_0}\right)\vec{\mathbf{B}}_0 R^3}$$

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